

## Review article

## Socio-ecological context and tree care for street and park trees in urban and suburban areas

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## ABSTRACT

Street and park trees, when growing in some of the most stressful conditions in urban areas, are important parts of the urban forest but often have low annual survival rates. Tree care, whether performed by public agencies, private companies, or residents, can support tree health, growth, and associated ecosystem services. Urban tree research provides the opportunity to situate tree care practices in their urban socio-ecological context (i.e. land use, neighborhood, climate, planting site type) and test how that context impacts tree care outcomes. Here we perform a literature review to characterize the academic literature's research on whether and how the effectiveness of tree care interventions vary not just based on tree characteristics but also with the socio-ecological context. We find that some practices have been effective in different cities and land uses including pest management, mulching with compost, watering, and use of paclobutrazol to control or ferric ammonium citrate to treat chlorosis. Other practices are studied in different cities but don't seem as effective, like fertilization with just nitrogen. For practices like pruning and watering, the effectiveness seemingly varies depending on where and when the practice is tested. It was challenging to characterize how socio-ecological context impacted tree care effectiveness since several elements of especially the built environment and social context are rarely described, though other elements are commonly described, like tree care history, climate, and some information about the soils and planting methods. Even for relatively effective tree care practices, more effort is needed to better understand how the socio-ecological context, especially social conditions, impact their effectiveness.

## 1. Introduction

City trees provide a wide array of ecosystem services and supporting their health is vital for broader urban resilience and health. The urban forest as a whole contributes ecosystem services including urban heat island mitigation, stormwater management, pollution reduction, habitat, and supporting human mental and physical health (Livesley et al., 2016). For city trees to provide their full benefits, they need to survive to maturity and remain healthy (Turner-Skoff and Cavender, 2019). Yet, when and how tree care practices support tree health and ecosystem services across heterogeneous city environments is not well

established in the scientific literature. Trees on public property present an opportunity for city and regional governments, non-profits, and residents to collaborate in tree care. However, this collaborative care performed by diverse actors also means there is a social network, in addition to the physical heterogeneity in planting site type, temperature, and other conditions that vary across cities, that influences the type of care and the underlying growing conditions a tree experiences. Understanding whether the academic literature studies how this underlying social and physical heterogeneity influences the effectiveness of tree care for public trees can work alongside current best management practices to inform government policy, cross-sector collaborations, and

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tree planting and care practices.

Here, we compile studies performed in cities from the academic literature that investigate the impacts of tree care on street and park tree health, taking note of when and how the research examines relevant socio-ecological conditions (e.g., soil type, climate). There are a wide array of resources available on urban tree care provided by professional societies (e.g., the International Society of Arboriculture and Tree Care Industry Association), private companies, academic institutions, arborists, and the individual expertise of others who work with urban trees. Though they are supported by extensive experience and often rigorous research, these sources are often disparate, and the lack of synthesis can hinder policy and investment decisions. Publication in the academic literature is a method used to share information on the effectiveness of tree care practices across cities and is the focus of this particular review. Synthesizing how and whether socio-ecological conditions are described in the academic literature can suggest opportunities for further research to complement the embedded experience of tree care professionals that informs how we choose tree care approaches to use in a given socio-ecological context.

Reviews on tree care practices and their cost effectiveness have been conducted (e.g. Clark and Matheny, 2010; Sæbø and Ferrini, 2006; Vogt et al., 2015a), and so have reviews that discuss the importance of socio-ecological context for urban forestry (e.g. Vogt, 2024). This study aims to tie these two concepts together, using the tree care research performed in urban areas to qualitatively assess our academic understanding of how socio-ecological context influences the effectiveness of tree care practices across urban areas. The socio-ecological contexts are not always clearly described in the academic literature, and it is important to understand where/if urban conditions may render a practice more or less effective (Vogt, 2024). Because urban areas are heterogeneous, it is important to also understand the broader socio-ecological context of a study and clarify when a study tests the importance of the context beyond it being urban (e.g. construction history, site type, land use, resident involvement). Trees are also not homogenous in their tolerance of urban conditions and their tree care needs (Vogt et al., 2015a; Percival, 2023), so tree species and tree life stage are similarly important for understanding study applicability. This paper addresses these gaps by compiling papers on urban tree care, providing a resource alongside general arboricultural best management practices for those looking to see what tree care practices are effective for specific trees in a given context. Additionally, the paper notes areas that would benefit from further study or translation from knowledge within the professional industry.

### 1.1. Urban growing conditions and tree care

Arboriculture, horticulture, silviculture, and forestry have all contributed to our understanding of how growing conditions influence tree health, growth, and survival as well as how tree care or stand management practices can modify those outcomes (Hauer et al., 2017). Researchers from the government, private sector, and universities have for decades developed best management practices in arboriculture and advanced our understanding of tree biology and how it influences tree risk in urban contexts (e.g. Funk, 1988; Nielsen, 1989; Shigo and Marx, 1977; Smiley et al., 2000; Hauer and Koeser, 2018; Hargrave et al., 2023). Erik Jorgensen from the University of Toronto coined the term “urban forestry” in 1965 to focus explicitly on trees growing within a heavily human-modified, urbanized landscape as urban forestry became a discipline (Jorgensen, 1986; O’Herrin et al., 2023). Our understanding of the effectiveness of tree care practices comes from the experience of practitioners alongside decades of research within urban forestry as well as an even longer tradition of research within other related fields (e.g. Clark and Matheny, 2010; Hauer and Koeser, 2018).

For this paper, we define tree care to encompass all actions intentionally performed by people to support tree health and provision of ecosystem services after a tree is planted. We will use the terms

*stewardship* and *maintenance* to refer to complementary tree care activities. We define stewardship as activities that support tree health and ecosystem services that are relatively accessible to a non-professional. Relatively accessible here means that it does not require expensive or hard-to-use equipment and could be done by a layperson with some minimal training. Stewardship is particularly common for small, young trees, and examples of stewardship include mulching, watering, and pruning of small trees. Maintenance, on the other hand, includes tree care activities that require technical expertise or complex equipment. These specialized interventions are typical of the later years of a tree’s life. Examples of maintenance include pruning large trees (i.e. with tree climbing or lifts), root remediation, and some pest management.

Tree selection often hinges on the idea of “right tree, right place”, which requires an understanding of individual tree needs, the variable impact of urban growing conditions across species, and the needs and interests of local communities (Breyer and Mohr, 2023; Morakinyo et al., 2020). Despite extensive work to develop standards and practices related to tree care, there remains a lot of uncertainty about the effectiveness of practices like tree site selection, soil remediation efforts, and other site-specific elements to tree care (Vogt et al., 2015a; Vogt, 2024). For example, while nitrogen fertilization can have benefits for shade trees (Struve, 2002), it may have low importance in urban contexts (Harris et al., 2008; Rich and Walton, 1979; Werner and Jull, 2013). A combination of experimental trials in controlled environments and field trials embedded in urban street and park environments would clarify which tree care practices are effective for street and park trees and when they are most effective (Vogt, 2024).

Research performed in forested, more rural settings may not apply as well to trees in heavily built environments due to the unique stress of urban site conditions and the individualized tree care that street and park trees can receive. Street and park trees are often first grown in nurseries and then transplanted into the city. Nursery conditions can influence tree survival (Allen et al., 2017) but transplanting is stressful for trees and impacts tree growth for at least 6 months and often much longer (Watson, 2005). Trees are particularly vulnerable in this period soon after transplanting and may require frequent stewardship (Hilbert et al., 2019; Vogt et al., 2015a). As trees mature, they are able to provide a range of ecosystem services and some of the stewardship costs may go down as they establish and recover their root growth (Vogt et al., 2015a; Watson, 2005). However, trees have different growth and survival rates in comparatively stressful street environments as compared to more rural contexts, which need to be planned for (Smith et al., 2019). Tree needs and vulnerabilities vary across tree sizes and ages and therefore appropriate tree care changes throughout a tree’s life (Vogt et al., 2015a). Understanding the importance of urban growing conditions across a tree’s lifespan can further inform tree care practices to support tree growth and health.

Because the urban forest includes trees growing in a wide range of conditions, it would be difficult to compile effective tree care practices for all these conditions within a short paper. Street and park trees are often managed at the scale of an individual tree rather than a stand, which requires stewardship and maintenance practices that may not be as relevant to larger stands of trees (Piana et al., 2021). Trees on public property present an opportunity for collaborative stewardship because city and regional governments overlap with care provided by a mix of nonprofits, private businesses, and local residents. Street trees also grow in particularly stressful conditions, often with extensive pavement over the soil, in competition with a range of other infrastructure (e.g., buildings, utility lines, sewage, etc.), and near roadways with high levels of some pollutants (Czaja et al., 2020; Roman et al., 2020; Smith et al., 2019). Unsurprisingly, street trees tend to have high mortality rates (Hilbert et al., 2019; Smith et al., 2019). Trees with low health and/or that die quickly provide limited ecosystem services and require replacement tree and installation costs (Roman et al., 2020; Vogt et al., 2015a). Thus, improving and supporting the health of street trees and public park trees in heavily urbanized areas is particularly important for

creating equitable and robust urban green infrastructure.

As socio-ecological systems, urban forests are impacted by a range of social, cultural, economic, and political processes that also impact human communities (Heynen et al., 2006; Riley and Gardiner, 2020; Vogt, 2024). Street trees and park trees are particularly embedded in complex socio-ecological systems because they exist in particularly heavily built-up environments and are managed by a mix of public agencies and, sometimes, private businesses, residents, resident associations, and nonprofits (Breger et al., 2019; Czaja et al., 2020; Roman et al., 2020; Smith et al., 2019). The degree to which the current academic tree care literature is attentive to these influences is unclear. If the academic literature does not account for the importance of relevant social, cultural, economic, or political variables and processes, results may not reflect the varying needs of trees across urban conditions.

## 1.2. Research question

Using the academic literature, we focused on aspects of tree stewardship and maintenance that are relatively well-studied for urban street and park trees. Our aim was to understand the impacts of stewardship and maintenance on tree health. Specifically, our main research question was: from the academic literature, how much and what do we know about the effectiveness of stewardship and maintenance in supporting tree health, growth, and ecosystem services? We then explored three interrelated sub-questions:

1. How does socio-ecological context inform the effectiveness of stewardship and maintenance practices? What elements of the socio-ecological context are consistently mentioned in the academic literature on tree stewardship and maintenance? Which are actually tested?
2. Is there a bias in the literature towards any specific tree life stages? Are there forms of stewardship and maintenance that are more effective for specific tree life stages?
3. Which stewardship and maintenance practices have robust support in the academic literature and which less so? What do we know about the effectiveness of stewardship and maintenance across contexts from well-studied practices?

## 2. Methods

We constructed our research question using the PICO framework (Richardson et al., 1995) to collate studies of tree care that were done in cities or simulated urban growing conditions. PICO stands for Population (plain text), Intervention/Exposure (**bold**), Comparator (underline), and Outcome (*italics*), which can be mapped on our focal search question as follows:

How do **stewardship and maintenance (including mulching, pruning, watering, pest management, and structural support)** impact street and park tree *health, growth, and survival* in built environments (especially cities) compared to trees that receive less or no maintenance/stewardship?

### 2.1. Search string and scoping

We used a consistent search string to find relevant papers, adapted only for individual variations in syntax and word limits in the search function of individual databases. We started with 10 papers we knew we wanted our search string to pick up, drawing on the personal expertise of authors to add in forms of stewardship and maintenance that were of particular interest (Blair et al., 2019; Breger et al., 2019; Himelick and Watson, 1990; Lecigne et al., 2018; Luley et al., 2002; Roman et al., 2015; Ryder and Moore, 2013; Takamatsu and Yarita, 2020; Tan and Shibata, 2022; Vogt et al., 2015b). Then we iteratively added and removed keywords to see if any added a large number of irrelevant papers or added relevant papers. We did a similarly iterative process to

add and remove exclusion terms to help remove irrelevant papers while keeping relevant ones (Foo et al., 2021). The final search string is approximately as follows (though see [supplement Table S.1](#) for the full list of the search strings used for each database):

*(compost\* OR mulch\* OR prun\* OR maint\* OR care OR "pest manag\*" OR water\* OR support\* OR steward\*) AND ("street tree\*" OR "park tree\*") AND (health OR growth OR mortality OR function\* OR "ecosystem service\*") AND (city OR urban OR municipal\*) NOT ("decision tree" OR "classification tree" OR agricult\* OR "national park" OR "tree algorithm")*

Around March 30th, 2022, we ran the search in six databases: Web of Science, SCOPUS, CAB Abstracts, Environment Complete, ProQuest Dissertations and Theses Global, and the Science.gov portal. We explicitly included dissertations and theses alongside white papers and book chapters where relevant. We also pulled search results from the first 10 pages of Google Scholar results and hand-searched Arborist News, the Metropolitan Tree Research Alliance (METRIA), and TREE fund research. To be confident that our search would yield the original research papers that described the study site conditions, we performed a backward search (screening all the referenced articles) from six relevant review articles (Raupp et al., 1999; Sæbø and Ferrini, 2006; Roy et al., 2012; Ferrini and Fini, 2013; Vogt et al., 2015a; Hilbert et al., 2019). As part of the peer review process, a reviewer suggested that we pay particular attention to the journal *Arboriculture and Urban Forestry*. Although it is indexed in one of the databases we searched, we conducted an exhaustive secondary search of the journal using multiple individual keyword searches on October 26, 2024, to account for the characteristics of the article search on the journal's website (see [supplement Table S.2](#) for all terms used). We included all papers in either English or Spanish, limited by the knowledge of the screeners, though the search strings used were in English.

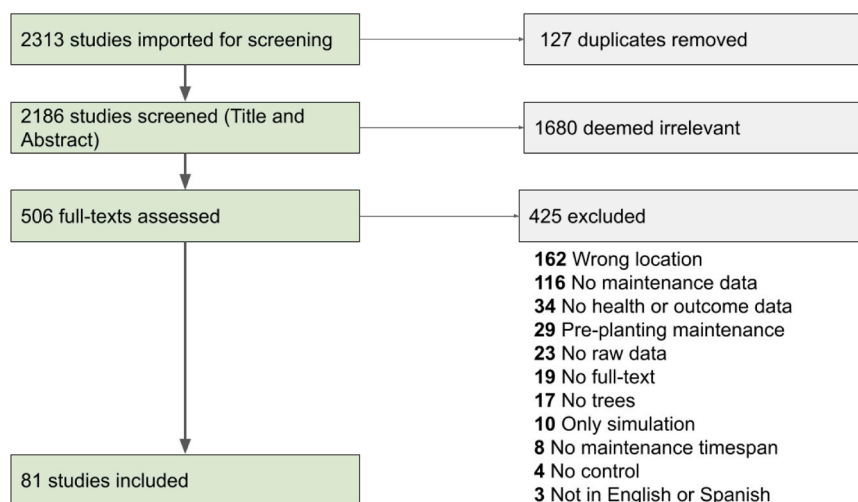
The initial search also allowed for papers that focused on outcomes like ecosystem service provision and tree function, hence the inclusion of both terms in the initial search string. However, the number of papers that discussed ecosystem services was very low, likely due to the fact that many ecosystem service papers name the specific service rather than using the general term (e.g. cooling, stormwater management). Therefore, here we focus on papers that highlight tree health, growth, and survival as outcomes.

### 2.2. Screening

Our search yielded 2186 unique papers (see [Fig. 1](#) for number of texts screened), which were exported to Covidence for screening. To ensure consistency, the same two screeners (both co-authors) screened the same 100 papers and then compared results, discussing disagreements to come to a consensus. Screeners were part of the creation of the protocol and discussed it with a fellow co-author with extensive library and systematic review experience before beginning the initial screening as well as throughout the screening process to ensure both understood the protocol and could implement it consistently. They then screened another 50 papers to ensure there was more than 90 % agreement. After that, papers were screened by title and abstract by one of the two screeners. A similar process was then used for the full-text review and for the later search of *Arboriculture and Urban Forestry*.

Our inclusion and exclusion criteria required that:

- Trees must be street or park trees, or the growing conditions must be set up to emulate those conditions.
- Data must be collected in a built environment or in conditions set to emulate a built environment (city, town).
- The papers must investigate numerous levels of tree care intensities and/or have at least a tree care treatment and a control treatment. Tree care in this case only includes actions that happen after the tree has been planted and before the tree is removed.
- Health/growth/survival data must be gathered and tied to tree care data.



**Fig. 1. PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) diagram for search results.** The diagram shows the flow of the literature screening process, including how many studies were removed or advanced to the next stage (Moher *et al.*, 2009). Reasons studies were excluded from the final review, and the number of studies for each reason are detailed beneath the “excluded” box.

- Full text must be available in English or Spanish (due to limitations of author expertise).

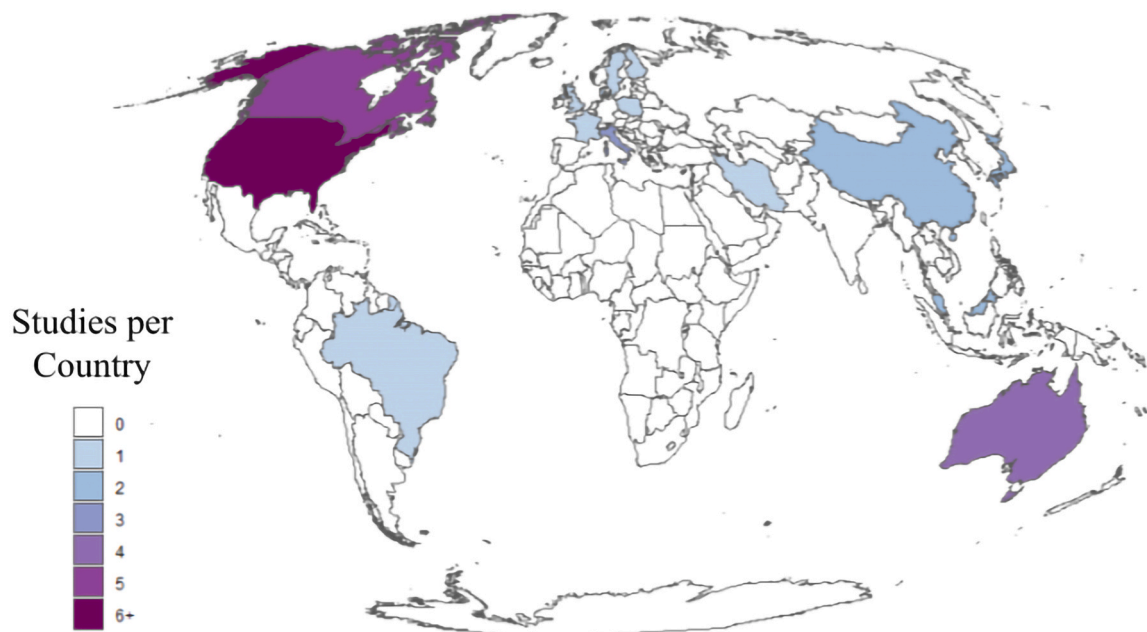
### 2.3. Data extraction

To get information from the papers that made it through the full-text review, we relied on a template from the platform Covidence (Covidence, 2023). We extracted data on the species and life stage of the trees investigated, the location, elements of the socio-ecological context mentioned, and the impact of maintenance/stewardship investigated. We also described any inconsistencies or lack of clarity in the paper methods, analysis, and extrapolations from analytical results as a data quality assessment.

We split tree life stages into roughly four categories: 1) establishment, which is less than 5 years post-planting, 2) juvenile, which is 5–10

years post-planting or growing but still not at mature size 3) mature, which in this case is any tree that is at least 10 years post-planting or has reached what is approximately the full size of the tree and 4) mixed, which is some mix of mature and either establishment or juvenile. The point at which a tree is considered mature varies based on species, intended use, and site conditions, but we use 10 years as a convenient cutoff for consistency’s sake. Where year post-planting is not available, we compare listed tree sizes to what is considered “mature” size for the species. For studies where tree age and size are not clearly listed, we simply put unknown (12 studies).

To create a list of elements of the socio-ecological context, first all the papers were read once, noting what elements of the context were discussed. Then, the elements noted were grouped to come up with general categories of information discussed in the papers, compiling a list of elements to look for (see [supplement Table S.5](#) for the final categories).



**Fig. 2. Geographic distribution of papers included in study.** The total number of papers included is 81, and 54 were from the United States (here listed as 6+). The map was created using the *rnaturalearth*, *sf*, and *ggplot2* packages in R (Wickham, 2016; Massicotte and South, 2023; Pebesma and Bivand, 2023), with a mollweide projection.



Each paper was then read again, looking both for whether a given element was mentioned at all in the paper and whether it was tested in the study.

### 3. Results

Our screening resulted in 81 final studies included in this analysis (Fig. 1), the majority of which were performed in the United States (Fig. 2) and were in the journal *Arbiculture & Urban Forestry* (formerly *Journal of Arbiculture* - supplementary Fig.S.1). Similarly, most species that were studied in more than one paper are native to North America (supplementary Table S.3). The studies spanned a range of tree life stages (Fig. 3). Around a third of studies were on a mix of tree life stages (25 papers), and a similar numbers looked at just mature trees (13) compared to just tree establishment (10) or juvenile trees (10 studies) though some looked at trees for long enough to see them transition from establishment to juvenile (4 studies) and some of the descriptions only provided tree sizes rather than ages, which vary with growing conditions along with species (e.g. Vaz Monteiro et al., 2017). Because of the overlap and uncertainty around tree age, from here on establishment and juvenile trees will be grouped together. The remaining studies did not report tree age or size (19 papers, Fig. 3). Though the tree ages included did vary by maintenance type (see below), overall, the results collated here are relevant to a range of tree ages.

There was a range of tree care practices present in the papers, and over a quarter investigated two or more forms of tree care simultaneously (22 out of 81). This allows for some understanding of how forms of stewardship and maintenance interact. Stewardship practices included: compost addition, general stewardship, mulching, pruning, tree guards, and watering (Table 1). Maintenance practices included: fertilization, foliar treatments, growth treatments, lintel (concrete) design, pest management, tree management policy, pruning, root barriers or excavations, staking, surface cover, tilling, and trenching (Table 1).

Most forms of maintenance or stewardship were only studied by a few studies (often fewer than four), and they were set primarily in the United States (Fig. 2). Because results are likely to vary depending on the context, we describe all final included papers in Table 1 and encourage interested readers to look at the original sources for the quantitative results.

#### 3.1. Socio-ecological context

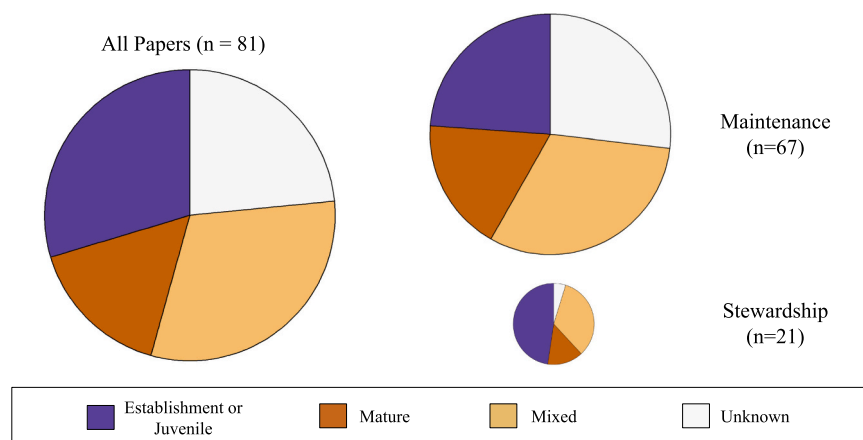
A broad range of variables related to the socio-ecological context were described, ranging from tree-specific to biophysical to social. Most studies described some of the relevant context of the study in their methods and introduction (Fig. 4, supplementary Table S.5). Tree-level information, particularly tree species and tree size or age, were described in the majority of studies (62 and 53 studies respectively) and were by far the most consistently described elements of conditions that could impact study outcomes. In slightly under half of studies, the climate was described, including temperature (25 studies), and precipitation (24 studies) along with season or time of year (32 studies). Some other elements of the growing conditions relating to soils and planning were described, including: land use (34 studies), site type (32 studies), location within the city (24 studies), city context within the region (24 studies), and soil type (23 studies). Descriptions of the urban forest broadly and initial tree health were also common. A general description of the urban forest was given in 23 studies, with tree care history (whether maintenance or stewardship) described in 31 studies. Tree disease history and initial health were described in 28 studies, though it was proportionally more common with studies looking at pest management (13 out of 23 papers).

In contrast, some of the aspects of the built environment were rarely described (Fig. 4). Building height or age, street orientation, and percent impervious surface cover were some of the more rarely mentioned aspects of the site information (four, seven and seven studies respectively), along with the development history of the city or area of interest (six studies). The social and socio-demographic context were also rarely described, with census or other socio-demographic information only given in eight studies and resident involvement described in 12.

Few studies were able to test the importance of the socio-ecological context, especially as it relates to the social context, soil conditions, and elements of the street and built environment (Fig. 4, supplementary Table S.5). That said, the impact of species or tree type and tree size or age were more regularly tested (in 36 and 25 studies respectively).

#### 3.2. Tree care across tree life stages

There was variation in the tree life stages studied depending upon the type of tree care, though many tree care categories had a mix of tree life stages represented (supplementary Table S.4). However, when splitting the tree care into stewardship (accessible to a layperson with minimal training or equipment) and maintenance (requiring technical expertise



**Fig. 3. Overall tree life stages investigated across stewardship and maintenance practices.** Tree age classes are as follows: 1) Establishment - less than 5 years post-planting and is just starting to grow again after transplant shock, 2) Juvenile - 5–10 years post-planting or growing but still not at mature size 3) Mature - any tree that is at least 10 years post-planting or has reached what is approximately the full size of the tree. Mixed connotes studies that look at multiple life stages. Unknown connotes studies that did not specify life stage. Note that the number of papers in the stewardship and maintenance categories do not add up to the total number in the “All Papers” category because papers could cover both stewardship and maintenance practices (n = 7).

**Table 1**

**Reference list of all studies included in the review.** Citation is listed along with tree care practices studied, tree life stages studied, and the location of the study. Papers are in alphabetical order by the author's last name.

| Citation                      | Tree care practices                                                                                                                                                                                            | Tree care type | Tree life stages                    | Location                                                 |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-------------------------------------|----------------------------------------------------------|
| Almas and Conway, 2018        | Urban Forest Management Plan                                                                                                                                                                                   | Maintenance    | Unknown                             | London, Hamilton, Oakville, and Markham, Ontario, Canada |
| Arron et al., 1997            | Growth Regulator Treatments (paclobutrazol, flurprimidol, dikegulac, uniconazole)                                                                                                                              | Maintenance    | Unknown                             | California and Nevada, USA                               |
| Ball, 1992                    | Pest Management (Bronze Birch Borer), Pruning (general, with dressing)                                                                                                                                         | Maintenance    | Juvenile                            | Duluth, Minnesota, USA                                   |
| Barger and Cannon, 1987       | Pest Management (European Elm Bark Beetles), Pruning (general, with dressing)                                                                                                                                  | Maintenance    | Unknown                             | Shaker Heights, Ohio, USA                                |
| Blair et al., 2019            | Watering, Staking                                                                                                                                                                                              | Both           | Establishment                       | Florida, USA                                             |
| Boyce, 2010                   | General Stewardship                                                                                                                                                                                            | Stewardship    | Mixed                               | TriBeCa, New York City, New York, USA                    |
| Braga et al., 2020            | Pruning (general)                                                                                                                                                                                              | Maintenance    | Unknown                             | Belo Horizonte, Brazil                                   |
| Breger et al. 2019            | Watering; General Stewardship                                                                                                                                                                                  | Stewardship    | Establishment                       | Holyoke, Massachusetts, USA                              |
| Bühler et al. 2006            | Watering                                                                                                                                                                                                       | Stewardship    | Juvenile                            | Frederikssundsvej, Copenhagen, Denmark                   |
| Cannon and Worley, 1976       | Pest Management (Dutch Elm disease)                                                                                                                                                                            | Maintenance    | Mixed                               | USA (various cities)                                     |
| Chase et al. 2021             | Pest Management ( <i>Melanaspis tenebricosa</i> )                                                                                                                                                              | Maintenance    | Juvenile                            | Rock Hill, South Carolina, USA                           |
| Coombes et al. 2019           | Pruning (crown lift, crown reduction or thinning, combinatorial pruning)                                                                                                                                       | Maintenance    | Mixed                               | Norwich and Peterborough, UK                             |
| Cregg and Ellison-Smith, 2020 | Growth Regulator Treatments (Paclobutrazol)                                                                                                                                                                    | Maintenance    | Mature                              | Lansing, Michigan, USA                                   |
| Doccola et al. 2011           | Pest Management (general)                                                                                                                                                                                      | Maintenance    | Mature                              | East Lansing, Michigan, USA                              |
| Faccoli and Gatto, 2016       | Pest Management (Asian Longhorned Beetle)                                                                                                                                                                      | Maintenance    | Mixed                               | Cornuda, Italy                                           |
| Fahrner et al. 2017           | Pest Management (Emerald Ash Borer)                                                                                                                                                                            | Maintenance    | Mixed                               | Minneapolis St. Paul, Minnesota, USA                     |
| Ferrini and Baietto, 2006     | Fertilizer (Nitrogen)                                                                                                                                                                                          | Maintenance    | Establishment                       | Milan, Italy                                             |
| Fini et al. 2016              | Mulch, Compost                                                                                                                                                                                                 | Stewardship    | Establishment                       | Florence, Italy                                          |
| Follett et al. 2016           | Pruning (heading cuts, directional pruning)                                                                                                                                                                    | Maintenance    | Mixed                               | Montreal, Canada                                         |
| Fraedrich and Ham, 1982       | Mulch                                                                                                                                                                                                          | Stewardship    | Juvenile                            | Clemson and Aiken, South Carolina, USA                   |
| Funk and Peterson, 1980       | Fertilizer (Manganese)                                                                                                                                                                                         | Maintenance    | Unknown (likely Juvenile or Mature) | Grand Rapids, Michigan, USA                              |
| Ghani et al. 2009             | Trenching                                                                                                                                                                                                      | Maintenance    | Unknown                             | Kuala Lumpur, Malaysia                                   |
| Gill et al. 1992              | Pest Management (Peachtree borer)                                                                                                                                                                              | Maintenance    | Establishment                       | Fairlakes, Virginia, USA                                 |
| Gilman, 2006                  | Root barriers                                                                                                                                                                                                  | Maintenance    | Juvenile                            | Gainesville, Florida, USA                                |
| Goodfellow et al. 1987        | Pruning (utility line clearance)                                                                                                                                                                               | Maintenance    | Unknown                             | Green Bay, Wisconsin, USA                                |
| Green and Watson, 1989        | Mulch, Tilling                                                                                                                                                                                                 | Both           | Juvenile                            | USA (specific location unclear)                          |
| Gregory and Allison, 1979     | Pest Management (Dutch Elm disease)                                                                                                                                                                            | Maintenance    | Unknown                             | Shaker Heights, Ohio and Elmhurst, Illinois, USA         |
| Großmann et al. 2020          | Pruning (Powerline pruning)                                                                                                                                                                                    | Maintenance    | Mixed                               | Montreal and Quebec, Canada                              |
| Harrell et al. 1984           | Fertilizer (Nitrogen, Phosphorus and Potassium; High Phosphorus), Foliar Treatment (Medi-Ject; Medicap FE/MN/MD; Zinc/Iron-Zinc; Iron Sulfate; FeHDTA; FeSO <sub>4</sub> + S; MnSO <sub>4</sub> , Citric Acid) | Maintenance    | Mixed                               | Lincoln and Omaha, Nebraska, USA                         |
| Harrell et al. 1988           | Foliar Treatment (Ferric Ammonium Citrate, NPK injections)                                                                                                                                                     | Maintenance    | Mixed                               | Omaha, Nebraska, USA                                     |
| Harris et al. 2008            | Fertilizer (Nitrogen)                                                                                                                                                                                          | Maintenance    | Establishment                       | Blacksburg, Virginia USA                                 |
| Hartman et al. 2010           | Growth Treatment (Paclobutrazol), Pest Management (bacterial leaf scorch)                                                                                                                                      | Maintenance    | Mature                              | Lexington, Kentucky, USA                                 |
| Haugen et al. 2016            | Pruning (utility line clearance), Growth Treatment (Paclobutrazol)                                                                                                                                             | Maintenance    | Unknown                             | Baton Rouge, Louisiana, USA                              |
| Himelick and Himelick, 1980   | Foliar Treatment (Iron), Fertilizer (Nitrogen, Phosphorus, Potassium)                                                                                                                                          | Maintenance    | Mixed                               | Urbana, Illinois, USA                                    |
| Himelick and Watson, 1990     | Mulch, Foliar Treatment (ammonium sulfate)                                                                                                                                                                     | Both           | Mixed                               | Lake Forest, Illinois, USA                               |
| Holmes and Davidson, 1984     | Pest Management (general)                                                                                                                                                                                      | Maintenance    | Unknown                             | Washington DC Metropolitan Region, USA                   |
| Jack-Scott et al., 2013       | General Stewardship                                                                                                                                                                                            | Stewardship    | Mixed                               | New Haven, Connecticut, USA                              |
| Karlovich et al. 2000         | Pruning (Topping)                                                                                                                                                                                              | Maintenance    | Unknown                             | Illinois, USA (various cities)                           |
| Kjelgren and Clark, 1993      | Watering, Fertilizer (Nitrogen, Phosphorus, Potassium)                                                                                                                                                         | Both           | Mature                              | Seattle, Washington, USA                                 |
| Lawson and Dahlsten, 2003     | Pest Management (Elm Leaf Beetle Monitoring)                                                                                                                                                                   | Maintenance    | Mixed                               | Sacramento, California, USA                              |
| Lecigne et al. 2018           | Pruning (V-trimming)                                                                                                                                                                                           | Maintenance    | Mature                              | Ahuntsic district, Montreal, Canada                      |
| Lecigne et al. 2020           | Pruning (Side trimming, top trimming, V-trimming)                                                                                                                                                              | Maintenance    | Unknown                             | Quebec, Canada (11 cities)                               |
| Lin et al., 2021              | General Stewardship, Tree guards                                                                                                                                                                               | Stewardship    | Mixed                               | New York City, New York, USA                             |
| Lindén et al. 2020            | Mulch, Watering, Fertilizer, General Stewardship                                                                                                                                                               | Both           | Mixed                               | Helsinki, Finland                                        |

(continued on next page)

Table 1 (continued)

| Citation                   | Tree care practices                                                                    | Tree care type | Tree life stages                 | Location                                                       |
|----------------------------|----------------------------------------------------------------------------------------|----------------|----------------------------------|----------------------------------------------------------------|
| Luley et al. 2002          | Pruning (general)                                                                      | Maintenance    | Mixed                            | Rochester, New York, USA                                       |
| May et al. 2013            | Watering                                                                               | Stewardship    | Unknown                          | Melbourne, Victoria, Australia                                 |
| Morgenroth, 2011           | Surface Cover (Porous vs impervious pavement, gravel subgrade)                         | Maintenance    | Establishment                    | Christchurch, New Zealand                                      |
| Noack et al. 2009          | Pest Management ( <i>Thaumastocoris peregrinus</i> )                                   | Maintenance    | Mature                           | Sutherland, Australia                                          |
| Noukoun et al. 2014        | Pest Management (Cankerworm)                                                           | Maintenance    | Mature                           | Raleigh, North Carolina, USA                                   |
| Oldfield et al. 2015       | Mulch, Compost                                                                         | Stewardship    | Establishment                    | New York City, New York, USA                                   |
| Oleksyn et al. 2007        | Mulch, Fertilizer (Nitrogen, Sulfate)                                                  | Both           | Mature                           | Aleja Wielkopolska in Poznan, Poland                           |
| Omar et al. 2018           | Surface cover (equipment at tree base)                                                 | Maintenance    | Unknown                          | Bercy district, Paris, France                                  |
| Qu et al. 2019             | Mulch                                                                                  | Stewardship    | Mature                           | Haidian district, Beijing, China                               |
| Rathjens et al. 2009       | Root Crown Excavation                                                                  | Maintenance    | Juvenile                         | Cincinnati, Dublin, and Strongsville, Ohio, USA                |
| Raupp et al. 2001          | Pest Management (general)                                                              | Maintenance    | Mixed                            | Maryland, USA                                                  |
| Raupp et al. 2004          | Pest Management (Mites)                                                                | Maintenance    | Mixed                            | Washington-Baltimore Metropolitan Region, USA                  |
| Rich and Walton, 1979      | Fertilizer (Nitrogen)                                                                  | Maintenance    | Juvenile                         | Hamden, Connecticut, USA                                       |
| Roman et al. 2015          | Mulch, Pruning, Watering, Staking, General Stewardship                                 | Stewardship    | Establishment to Juvenile        | East Palo Alto, California and Philadelphia, Pennsylvania, USA |
| Roseli et al. 2021         | Growth Treatment (Paclobutrazol, Potassium Nitrate)                                    | Maintenance    | Juvenile                         | Kuala Lumpur, Malaysia                                         |
| Ryder and Moore, 2013      | Pruning (Formative)                                                                    | Maintenance    | Establishment - Juvenile, Mature | Melbourne, Australia                                           |
| Sadof et al. 2017          | Pest Management (Emerald Ash Borer)                                                    | Maintenance    | Mixed                            | Indiana, USA (three cities)                                    |
| Salahi et al. 2017         | Foliar Treatment (malic acid, citric acid, glutamine, sequestrane)                     | Maintenance    | Unknown                          | Tehran, Iran                                                   |
| Sand et al. 2018           | Surface cover (permeable vs impermeable ground cover)                                  | Maintenance    | Mature                           | Göteborg, Sweden                                               |
| Scharenbroch et al. 2022   | Fertilizer (biochar), Tilling                                                          | Maintenance    | Mixed                            | Bolingbrook, Illinois, USA                                     |
| Schultz, 1989              | Pest Management ( <i>Malacosoma disstria</i> )                                         | Maintenance    | Juvenile                         | Norfolk, Virginia, USA                                         |
| Sherald and Gregory, 1980  | Pest Management (Dutch Elm disease)                                                    | Maintenance    | Unknown                          | Washington D.C., USA                                           |
| Shogren and Paine, 2016    | Pest Management (Cuban Laurel Thrips)                                                  | Maintenance    | Unknown                          | California (various cities), USA                               |
| Smiley and King, 1998      | Foliar Treatment (Ferric Ammonium Citrate)                                             | Maintenance    | Unknown                          | Highland Park, Illinois, USA                                   |
| Švihra et al. 1999         | Staking (Double stack system, Bio-Tie, Tree Saver)                                     | Maintenance    | Establishment                    | Novato, California, USA                                        |
| Szota et al. 2019          | Lintels                                                                                | Maintenance    | Unknown                          | Melbourne, Australia                                           |
| Takamatsu and Yarita, 2020 | Pest Management (Fall webworms)                                                        | Maintenance    | Mixed                            | Hakodate City, Japan                                           |
| Tan and Shibata, 2022      | Pruning (general), Tree guards                                                         | Maintenance    | Mixed                            | Kyoto City, Japan                                              |
| Tate, 1980                 | Pruning (general), Fertilizer (Nitrogen, Phosphorus, Potassium), Remove Girdling Roots | Maintenance    | Mixed                            | Ann Arbor, Michigan, USA                                       |
| Ticehurst and Finley, 1988 | Pest Management (Gypsy Moth - <i>Lymantria dispar</i> )                                | Maintenance    | Unknown                          | Lake Barcroft, Virginia, USA                                   |
| Turner and Buss, 2005      | Pest Management ( <i>Allokermes kingii</i> )                                           | Maintenance    | Mature                           | Clearwater, Florida, USA                                       |
| Vogt et al. 2015b          | Mulch, Pruning, Watering, General Stewardship                                          | Stewardship    | Establishment - Juvenile         | Indianapolis, Indiana, USA                                     |
| Watson and Himelick, 2004  | Growth Treatment (Paclobutrazol, Flurprimidol)                                         | Maintenance    | Juvenile                         | Naperville, Illinois, USA                                      |
| Watson and Kupkowski, 1991 | Mulch                                                                                  | Stewardship    | Mixed                            | Chicago, Illinois, USA                                         |
| Werner and Jull, 2013      | Fertilizer (Nitrogen)                                                                  | Maintenance    | Mixed                            | Waukesha, Wisconsin, USA                                       |
| Whitlow et al. 1992        | Watering                                                                               | Stewardship    | Establishment                    | New York City, New York, USA                                   |
| Zhu et al. 2021            | Surface Cover (plastic blanket, tree grate, turf, wood board, cement, none)            | Maintenance    | Mature                           | Zhengzhou City, Henan, China                                   |

or equipment) a skew becomes more evident, as expected (Fig. 3). For stewardship practices, a plurality of papers looked at establishment or juvenile trees (10 of 21), with a little over a quarter looking at mixed groups of trees (7 of 21). For maintenance a plurality of papers looked at mixed tree life stages (21 of 67), with the next most focused on establishment/juvenile and then mature trees (16 and 12 papers respectively). For more details on tree life stages investigated for specific stewardship and maintenance practices, see [supplement Table S.4](#).

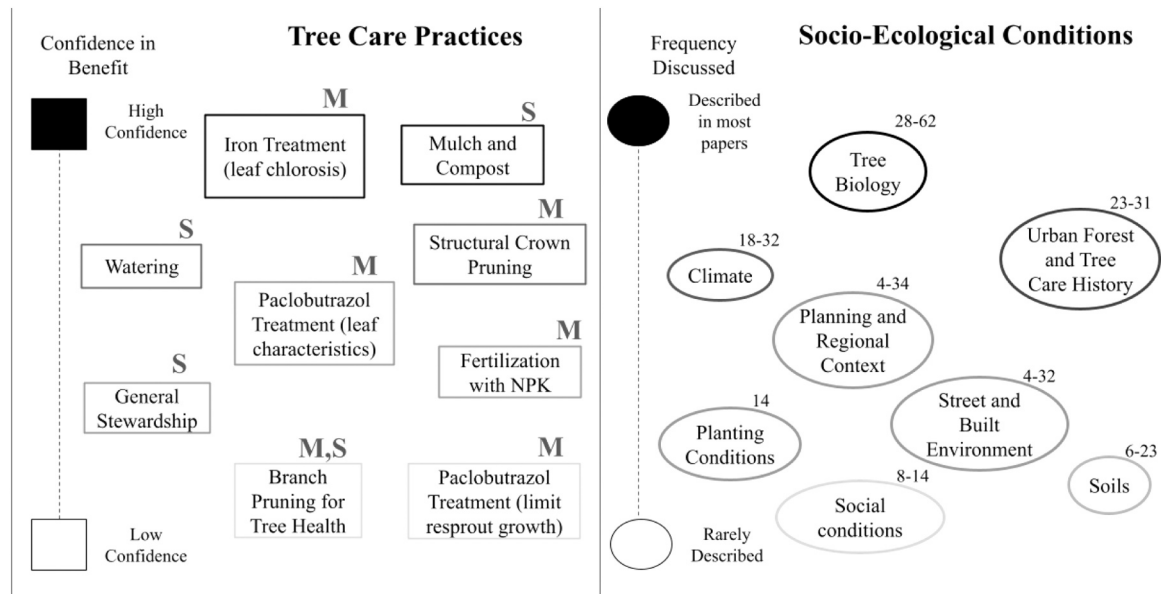
### 3.3. Impacts of tree care practices

For this discussion of how the effectiveness of tree care practices varies with context, we focus on stewardship and maintenance practices that were addressed by at least four studies to allow for some sense of how results might vary across contexts. This means that at least some of

the relevant results from 17 papers are not discussed here, but are summarized in the supplement. Ideally, we would discuss how tree care practices vary across contexts using more than four studies per tree care practice, but we were limited by the number of studies captured in our review. For a summary of the results for the other stewardship and maintenance practices, see the supplement. We encourage readers to look at [Table 1](#) for more context on the papers referenced below.

#### 3.3.1. Maintenance: short-term fertilization and nutrient treatments

Short-term impacts of fertilization with a combination of nitrogen, phosphorus, and potassium were often effective in increasing tree growth, though it was not as effective in remedying the impacts of girdling roots or symptoms of chlorosis. Fertilization with nitrogen, phosphorus, and potassium in combination (NPK;  $n = 6$  papers, 652 trees) had generally short-term positive or neutral impacts on tree



**Fig. 4.** Effectiveness of stewardship and maintenance practices and the description of socio-ecological conditions that may affect their benefits. This synthesis figure shows stewardship and maintenance practices (with evidence from more than four papers) and socio-ecological conditions that may impact their benefits. On the left is a scale from low confidence (light box borders) to high confidence (dark box borders) in the benefit of the practice. The letters next to the boxes signify what kind of tree care practice each example is, stewardship (S) or maintenance (M). The socio-ecological conditions (right) include a variety of factors that could impact the effectiveness of tree care practices. Analogous to the tree care practices, the darker the border, the more frequently those factors were discussed. The percentages to the top right of each oval show the ranges of how frequently each subtopic within these categories was described in the total of 81 papers included in the review. Note that, just because a socio-ecological condition was described, does not mean its importance was tested as part of the study methodology (see supplement Table S.5 for more details).

growth and health. In some cases, NPK led to higher growth rates for at least the first year after planting, though not always beyond (Ferrini and Baietto, 2006), though with mature trees it could increase growth more than irrigation alone for at least the growing season after treatment (Kjelgren and Clark, 1993), and could, within a growing season, improve the effectiveness of chlorosis treatments in some species (Himelick and Himelick, 1980). The benefits of NPK did vary by species (Ferrini and Baietto, 2006; Himelick and Himelick, 1980). However, in one study, 2 years of fertilization with NPK did not improve tree health or survival for trees with formerly girdling roots relative to controls (Tate, 1980). Fertilization with NPK also seemed to have limited impacts on leaf chlorosis. For *Quercus palustris* trees in two Nebraska cities, iron treatments provided the greatest improvement in leaf color, and trees only fertilized with NPK did not see different changes than control (Harrell et al., 1984). In the same region, a study of *Acer saccharinum* did not see an improvement in foliage color with NPK trunk injections, and again iron treatments were more effective (Harrell et al., 1988). Though not clearly effective in decreasing chlorosis symptoms, fertilization with NPK was generally effective at increasing tree growth but was less effective in correcting leaf chlorosis if used alone and appears to be less effective in Nebraska cities, though impacts vary by species.

### 3.4. Maintenance: paclobutrazol, growth, and foliar treatments

Due to an increase in branch regrowth after pruning (see below), paclobutrazol is sometimes deployed to limit resprout growth. Unfortunately, there is mixed evidence as to its effectiveness in reducing resprout (n = 4 papers, 301 trees). Arron et al. (1997) and Clegg and Ellison-Smith, 2020 both found that paclobutrazol had no clear impact on tree growth. Hartman et al. (2010) also did not find any changes in tree growth for *Quercus palustris* street trees, though there were some moderate reductions in diameter growth for trees in a golf course. However, Haugen et al. (2016) found for *Quercus virginiana* that application of paclobutrazol within 180 days pre or post pruning led to shorter branch lengths, less pruned biomass, and therefore shorter

pruning and chipping times three growing seasons after pruning, with even greater cost effectiveness when used within 90 days of pruning.

Paclobutrazol has also been used as a treatment for chlorosis or otherwise to boost tree tolerance to harsh urban conditions (n = 4 papers, 353 trees). For example, paclobutrazol did increase the chlorophyll index for *Pyrus calleryana* and *Gleditsia triacanthos* trees and increased net photosynthesis and stomatal conductance in a site with high impervious surface (Clegg and Ellison-Smith, 2020). Similarly, Roseli et al. (2021) found that treating *Xanthostemon chrysanthus* trees with paclobutrazol, with and without potassium nitrate, was associated with higher leaf chlorophyll content, reduced photosynthetic rates, and reduced stomatal conductance. Paclobutrazol also led to more compact leaf mesophyll and, with potassium nitrate, thicker palisade parenchyma (leading to greater leaf thickness - Roseli et al., 2021). Watson and Himelick (2004), using *Quercus palustris*, found that paclobutrazol led to a significant improvement in leaf color for chlorotic trees after four years and reduced color decline for non-chlorotic trees, though chlorosis intensity and the impact of paclobutrazol were more pronounced for street trees compared to trees on a golf course. However, when Hartman et al. (2010) tested the efficacy of paclobutrazol to manage bacterial leaf scorch in *Quercus palustris*, they found that street trees drenched with paclobutrazol saw the same increase in leaf scorch symptoms after a year as control trees. While there is mixed evidence of paclobutrazol's effectiveness in slowing regrowth, there is some evidence of its benefits for tree health, especially when used to treat chlorosis.

Iron supplements, like ferric ammonium citrate, are also sometimes used to address chlorosis symptoms (n = 4 papers, 727 trees). Ferric ammonium citrate and ferric citrate effectively corrected even some advanced stages of chlorosis in sweetgum and *Quercus palustris* but not in *Quercus alba* or *Platanus occidentalis* (Himelick and Himelick, 1980). Similarly, ferric ammonium citrate was effective in treating chlorosis in *Quercus palustris*, but not in *Acer saccharinum* for up to a year (Harrell et al., 1984), though it could be effective for *Acer saccharinum* when applied in the form of Iron NutriBoosters (Harrell et al., 1988).



Xylem-applied ferric ammonium citrate led to improvements in both *Quercus rubra* and *Quercus alba* chlorosis symptoms from iron deficiency as well, though these benefits lasted for four years (Smiley and King 1998). Ferric ammonium citrate treatments are often effective in treating chlorosis symptoms, though the length of effectiveness varies.

### 3.4.1. Maintenance: surface cover

There are differences in tree health depending on what kind of cover is near a tree base ( $n = 6$  papers, 4325 trees). *Platanus orientalis* saw more shallow root abundance, which can cause infrastructure damage, under pavements (both porous and impervious). However, root biomass was higher under porous pavements as compared to impervious (Morgenroth, 2011). Showing clearer signs of growth reduction under impervious surfaces, both stem growth and current-year shoot growth for mature *Tilia europea* in areas with less than 10 % permeable surface cover were lower than for trees with completely permeable ground surface cover (Sand et al., 2018). In Kyoto City, Japan, trees surrounded by flowerbeds, grass, or shrubs generally had higher health than those surrounded by concrete paving blocks or bare soil (Tan and Shibata, 2022). There tends to be higher species diversity at tree bases with non-compacted soils and animal excrement in Paris, France (Omar et al., 2018). A study of *Fraxinus chinensis* in Zhengzhou City, China saw a variety of differences in leaf properties (leaf thickness, chlorophyll index, leaf dry matter, leaf tissue density, specific leaf area, stomatal density, and stomatal area) depending on what tree pit covers were used. No one cover type maximized all the leaf properties measured and cement did not consistently rank the lowest (Zhu et al., 2021). In a New York City, US park, plots with shrubs had more tree stem volume and the more species-rich plots had higher basal area and height growth (Oldfield et al., 2015). Tree growth and health are impacted not just by whether the surface cover at the tree base is impervious versus pervious but more specifically by the type of impervious or pervious cover.

### 3.4.2. Maintenance: pest treatments

Because pest treatments are specific to the pest of interest and our search string did not include all potential species names (see supplementary table S.1 and S.2) here we discuss general takeaways of the challenges and opportunities within pest management programs.

Many papers tested chemical treatments as methods for controlling pest species, and all found they were effective in controlling the insect of concern ( $n = 8$  papers, 10,500 + trees; Chase et al., 2021; Doccola et al., 2011; Faccoli and Gatto, 2016; Gregory and Allison, 1979; Noack et al., 2009; Schultz, 1989; Sherald and Gregory, 1980; Turner and Buss, 2005). The one study that tried to manage a bacterial disease (bacterial leaf scorch) did not find chemical treatments to be effective (Hartman et al., 2010,  $n = 124$  trees). There are also known negative effects of insecticides and pesticides on urban ecosystems and human health (e.g. Md Meftaul et al., 2020), and some papers explicitly examined how sprays used to treat one pest can lead to increases in other pests due to unintended downstream consequences (Raupp et al., 2001,  $n = 26$  sites; Raupp et al., 2004,  $n = 59$  trees). Also, pruning that can be effective in controlling some diseases (like Dutch Elm Disease) can attract potentially damaging pests despite the use of dressings, and so are worth monitoring (Ball, 1992,  $n = 45$  trees; Barger and Cannon, 1987,  $n < 4021$  trees).

Integrated pest monitoring programs were one of the approaches that can reduce the chemical loads ( $n = 3$  papers, 11,500 + plants; Holmes and Davidson, 1984; Lawson and Dahlsten, 2003; Ticehurst and Finley, 1988). For example, Holmes and Davidson, 1984 estimated that an integrated pest management program for residential properties in the Washington D.C. area required slightly more person-hours due to monitoring needs but reduced the pesticide volume sprayed by 94 % across a season. Other larger pest management systems included tree monitoring and targeted removals or treatment for diseases like Dutch Elm disease (Cannon and Worley, 1976,  $n = 39$  sites), elm leaf beetle (Lawson and Dahlsten, 2003,  $n = 420$  trees), or emerald ash borer

(Fahrner et al., 2017,  $n = 2370$  trees; Sadof et al., 2017,  $n = 14,000$  + trees).

Other studied approaches included biological controls (Shogren and Paine, 2016,  $n = 11,000$  + trees; Gill et al., 1992,  $n = 20$  trees), or non-chemical traps like sticky bands or like komo traps (Noukoun et al., 2014,  $n = 33$  trees; Takamatsu and Yarita, 2020,  $n = 421$  trees). All the tested approaches were effective, but, like chemical controls, can also have unintended downstream ecological impacts and also require care for effective implementation (especially for biological controls; Heimpel and Cock, 2018).

### 3.4.3. Stewardship and maintenance: branch pruning

There is mixed evidence on the impact of pruning tree branches on tree health, growth, or survival depending on the pruning intensity and management objective ( $n = 5$  papers, 5000 + trees). Most of the studies on pruning looked at maintenance-level pruning (see supplementary Table S.4). For example, structural branch pruning early in a tree's life can correct for structural defects (like codominant stems) and save more than \$30 a tree in pruning costs when pruned at 3 and 7 years old compared to trying to correct the defects at 20 years old, accounting for inflation (Ryder and Moore, 2013). When connecting pruning to tree health, the included papers were less positive and some highlight potential challenges. Tate, 1980 found that pruning some live branches after removing girdling roots did not appear to affect foliage color, leaf size, crown diameter, or survival after two years. Similarly, when looking more at stewardship-level pruning, Vogt et al. (2015b) found that neither branch pruning nor whether the pruning followed best practices were included in the best model for tree growth rate or survival for trees in Indianapolis, U.S. Pruning can also attract potential pests like European elm bark beetles (Barger and Cannon, 1987) or bronze birch borer (Ball, 1992), despite the use of wound dressing. Tree branch pruning can be important for tree structural integrity, but can have more limited benefits for tree foliage color and can attract potential pest species.

The impacts of maintenance-level pruning on branch regrowth rates are similarly inconsistent ( $n = 4$  papers, 440 + trees). Lecigne et al. (2020) found that branch pruning around power lines did not appear to explain the growth of epicormic branching, a stress response, in most species besides *Quercus rubra*. However, Follett et al. (2016) found that tree branch growth tended to be greater immediately after pruning, though again results varied by species and type of branch pruning. Whether fast branch growth is desirable or not depends partly on which branches are growing, but Follett et al. (2016) was specifically measuring branches that regrew into the utility corridor and thus would need to be pruned out again. In that case, faster growth would lead to increased maintenance costs. Similarly, Goodfellow et al. (1987) found that regrowth rates in six species were highest in the season right after pruning and took more than four growing seasons to return to normal, which means that growth control (see above) would be most important soon after pruning. However, they also found that regrowth rates depend on the pruning type; side pruning and "natural pruning" (drop-crotch prune) had lower regrowth rates than round over and top pruning. Braga et al. (2020) found that intense branch pruning around power lines was loosely correlated with less branch regrowth when all species were pooled together. However, there were a wide range of species-level differences ranging from strong positive to strong negative correlations between pruning intensity and branch regrowth post-pruning (Braga et al., 2020). Follett et al. (2016), Goodfellow et al., 1987, and Braga et al. (2020) all indicate that the extent of species-level variation and the importance of the intensity and type of branch pruning means that the relationship between branch pruning and regrowth rates is heavily context-dependent.

Intense branch pruning can provide habitat for other species ( $n = 2$  papers, 2210 trees). Tan and Shibata, 2022 found that tree health condition was lower with intense branch pruning for all seven species investigated, though there were species-level differences. However,

Großmann et al. (2020) found that trees under power lines, receiving the most intense branch pruning, had similar microhabitat (e.g. holes for insects or birds) abundances to what you would see in a forest context, while park trees and street trees not under power lines had fewer microhabitats per tree. Many of these microhabitats in street trees under power lines were due to insect damage or other injuries, while the forest trees had more woodpecker cavities (Großmann et al., 2020). The results of these two studies are consistent with the idea that microhabitats in intense branch pruning scenarios often come from conditions that can negatively impact tree health.

#### 3.4.4. Stewardship: watering

Generally, irrigation had a positive impact on tree health and growth, though sometimes the impact was marginal ( $n = 6$  papers,  $n = 28,000 +$  trees). While some species seem to particularly benefit from dedicated irrigation systems, like palms, generally both dedicated irrigation systems and ad-hoc irrigation systems (i.e. with a water truck) correlated with improved tree quality and reduced dieback (Blair et al., 2019). When looking at trees partly stewarded by residents alongside nonprofits and government, tree irrigation plans that rely on a group of residents together (as opposed to individual residents) have higher survival in springtime, though may not be as effective in increasing growth, especially in the fall (Vogt et al., 2015b). There is evidence that irrigated trees have lower water stress as measured by stomatal conductance and predawn leaf water potential (Kjelgren and Clark, 1993; Whitlow et al., 1992). Irrigated trees may also have increased growth (Kjelgren and Clark, 1993) and potentially longer growing seasons (Bühler et al., 2006). However, the effectiveness of irrigation may vary by season (Vogt et al., 2015b) and species (May et al., 2013), and may be insufficient to satisfy tree water demand in extreme conditions, like drought in the summer (May et al., 2013). The way trees are irrigated, and the intensity of the underlying climatic conditions impact its effectiveness.

#### 3.4.5. Stewardship: mulch and compost

Mulching trees with organic mulch generally had positive effects on tree growth ( $n = 4$  papers, 841 trees, see supplementary Table S.6 for more details on mulch types). Correct mulching (i.e. not extensively piled) was positively correlated to growth rate in young trees in Indianapolis (Vogt et al., 2015b), and mulching with composted leaves and wood chips was positively correlated to growth in young *Acer saccharum* along a parkway (Green and Watson, 1989). Trees mulched with organic (wood chip) mulch may also have higher fine root development (Green and Watson, 1989; Himelick and Watson, 1990), though the difference for Himelick and Watson (1990) was not statistically significant. Fraedrich and Ham (1982) found that *Acer saccharum* and *Acer rubrum* mulched with wood chips grew both taller and wider at the sandy loam site and had higher current-year shoot growth on the clay loam site but not on the sandy clay loam site. Thus, the impacts of wood chip mulch on tree growth rates may depend on soil type and early season warming of soil from inorganic mulch (e.g. see Iles and Dosmann, 1999).

Compost or biochar were often studied as part of a mulching or fertilization scheme, and generally have positive impacts on tree growth and health, with potential negative impacts on mortality depending on the context ( $n = 5$  papers, 1498 trees). In one instance, the positive impacts of compost could not be teased apart from those of mulch, but trees with a mulch of wood chips and composted leaves were larger, greener, and less stressed (Green and Watson, 1989). Compost increased tree growth (Fini et al., 2016; Oldfield et al., 2015), and specifically mixed compost (with green material and kitchen waste) also increased leaf gas exchange (Fini et al., 2016). The combination of wood chip mulch and composted leaves also increased foliar N concentration, leaf mass, leaf area, and how long green leaves remained on the trees in *Aesculus hippocastanum* (Oleksyn et al., 2007). Though not exactly compost, Scharenbroch et al., 2022 ( $n = 75$ ) found that biochar, when combined with fertilization and tillage for 2 years, helped increase

growth rates for *Acer rubrum* trees, though they did not see similar improvements for *Gleditsia triacanthos* or *Ulmus parvifolia*. The impacts of compost may also be contingent upon various properties of the compost, which papers rarely reported; only Oleksyn et al., 2007 and Scharenbroch et al., 2022 provided nutrient concentration information (see supplementary Table S.6). However, the positive impacts of compost, at least on growth, were consistent across species though there was some variation in the degree of the increase in tree growth rates, and some species with high growth rates experienced increased mortality rates (Oldfield et al., 2015).

#### 3.4.6. Stewardship: general

The last category of papers are ones that look at the impact of stewardship generally rather than trying to disentangle the relative impacts of a variety of practices ( $n = 5$  papers, 600,000 + trees). Broadly, consistent and reliable stewardship increases the likelihood of tree survival, for both newly planted and established trees (Boyce, 2010; Breger et al., 2019; Roman et al., 2015). In one case, the positive impact of stewardship was tied to consistent watering, particularly during a drought (Breger et al., 2019). This stewardship can be led by non-profits or government agencies, but it requires communication, sufficient equipment, and labor, and often is aided by the support of volunteers (Boyce, 2010; Breger et al., 2019; Roman et al., 2015). In New Haven, Connecticut, Jack-Scott et al. (2013) found that the survival and growth of trees planted by community groups is related to group experience, longevity, and number of participants. However, success also depended on how closely the planting location aligned with a group's area of focus (e.g. groups focused on parks planting trees in parks) along with site type and elements of tree biology like species and age. Stewardship can improve tree survival when performed with consistency, effective coordination, and by groups working within their skillset.

### 4. Discussion

This review highlights that, though there are a wide array of stewardship and maintenance practices that can effectively increase street and park tree health, growth, and survival, understanding how that effectiveness varies with the socio-ecological conditions is difficult given the information presented in the papers reviewed here. Though tree characteristics in particular are generally well-described (including tree life stage) there is much less information on elements of the socio-ecological context related to social structures or even soil conditions. Even when comparing results from eight or more studies (as we did for fertilization for NPK) it is difficult to understand how the application of those standards may need to shift based on any given local context (i.e. why is NPK less effective than iron treatments in those two Nebraska cities specifically?). Therefore, attempts to use the literature to inform how tree care varies depending on tree species or tree size may be effective, but doing so based on institutional capacity or street size is much more challenging (Fig. 4). This reflects the challenge highlighted by recent work describing how even the robust set of best management practices and standards related to urban forest management still provide limited guidance on how to adjust or adapt approaches based on local context (Vogt, 2024).

#### 4.1. Sub-question 1 - urban socio-ecological context

The studies included in this review were conducted within urban or suburban contexts, thus providing the opportunity to look at the effectiveness of tree care practices within a complex socio-ecological system. However, because only some elements of the socio-ecological context are well-described (namely related to tree biology and the tree care history) it is challenging to draw inferences from multiple studies to understand the importance of socio-ecological context, especially for the social context and soil conditions. Understandably, few papers manage to test the impacts of even the well-described variables because of the

increased complexity of sampling numerous cities or experimentally manipulating many variables at once (e.g., sampling across soil types, species, land use, and resident involvement all at once requires a large sample size and complicated protocol to separate out each variable's impact). With the variables described, albeit rarely tested, in these papers, readers might have the information to compare results across contexts and account for differences due to species or tree size. Readers are, however, less likely to have sufficient information to look at the mediating influence of something like resident involvement in stewardship or other social patterns, even as important as they are for addressing patterns of inequality in cities (Heynen et al., 2006; Riley and Gardiner, 2020). From the few instances when papers looked at resident involvement in maintenance, for example, the effectiveness of tree planting and survival can depend on these social dynamics (Boyce, 2010; Breger et al., 2019; Jack-Scott et al., 2013; Vogt et al., 2015a). Other rarely described elements, like building height or age may have more limited effects on tree care, but there is evidence that building height can influence wind and light conditions that influence rates of tree failure (e.g. Manfra et al. 2022) and building age can influence the concentrations of contaminants like lead and other soil properties (e.g. Clark and Knudsen, 2013; Park et al., 2010). Because they are rarely documented, their potential impacts on tree care effectiveness are difficult to evaluate even when compiling information across studies. It is thus unclear how some elements of the socio-ecological contexts (like resident involvement or urban form) influence the effectiveness of the maintenance practices described here.

#### 4.2. Sub-question 2 - tree care across tree life stages

For maintenance practices, the papers here cover the range of tree life stages, while they skew towards younger trees for stewardship practices (Fig. 3). There is evidence that some tree care practices are particularly valuable for younger trees (e.g. structural pruning) and others for mature trees (e.g. pest management). However, there are a variety of practices that provide benefits throughout a tree's lifespan, like irrigation, mulch, or fertilization. There is evidence of benefits of both stewardship and maintenance, though there is relatively little research in the literature on the benefits of stewardship for mature trees. Stewardship and maintenance require different approaches as stewardship can pull in the involvement of residents and non-profits (Breger et al., 2019; Jack-Scott et al., 2013; Vogt et al., 2015b), while maintenance requires trained professionals and more extensive equipment (Haugen et al., 2016; Ryder and Moore, 2013). Both stewardship and maintenance support tree health, survival, and ecosystem services, but they may require different approaches to successfully implement.

#### 4.3. Sub-question 3 - effective stewardship and maintenance practices

While there is more evidence for some practices than others, the majority of stewardship and maintenance practices reviewed here have positive impacts on tree health, growth, or survival (Fig. 4). In particular, short-term mulch and compost application, irrigation, and ferric ammonium citrate treatments for chlorosis were well supported by the literature included in the review (Fig. 4). For fertilization (with compost or specific nutrients) in particular, we caution that there are a variety of potential long-term tradeoffs between rapid growth and tree survival and longevity (Oldfield et al., 2015; Brien et al., 2020), and the papers included here investigate impacts on trees for only a relatively short time window (< 2 years). Pest management as a general practice was well supported in the review, ranging from fully integrated pest management programs with extensive monitoring to the use of chemical, physical, or biological controls. Notably, integrated pest management programs could help reduce pesticide use and overall costs, and a few studies noted the potential unintended consequences of cover sprays of chemical controls and emphasize that they should be used with caution. Paclobutrazol also appeared to be relatively effective in influencing tree

leaf characteristics and potentially reducing chlorosis but had more mixed effectiveness in reducing tree regrowth rates. Further research looking at how to limit tension between trees and built infrastructure is particularly important as tree planting initiatives aim to increase tree canopy cover in cities. While early structural branch pruning did seem to be more cost-effective than later structural pruning, the majority of studies investigating the impacts of pruning on tree health did not see clear impacts, besides lower health with intense branch pruning and potential to attract mites or other pests.

#### 4.4. Limitations

It is likely not all papers relevant to this topic are included. Our search included papers English and Spanish, with other languages excluded. While a limitation, peer-reviewed papers in major journals are often published in English or with an English abstract, thus reducing this potential exclusion of some papers. Our search terms were designed to capture general tree care literature and so focused on general terms for practices (i.e. pruning) but there are numerous industry-specific terms that might be used in some of the literature (i.e. round over, reduction, trimming, topping, pollarding, coppicing). Many of the papers that use the more industry-specific terms in their titles still have the more general terms (pruning, in this case) in their keywords or abstracts. However, for pest management, there are likely papers on species considered "pests" that refer to them only by their species name that we would be more likely to miss. Thus, partly due to the logistics of creating a search string that is short enough to meet database limitations, it is possible some relevant papers were missed, especially for pest management. It is also possible that some papers were missed if the journal is not indexed or if data are not published formally with an accompanying report. To reduce this error, we hand searched some journals (e.g., Arborist News, Journal of Arboriculture and its new title Arboriculture and Urban Forestry), but there are likely some publications excluded, especially short summaries or overviews not published in the academic literature. Magazines and newsletters in particular are popular sources of information for arborists/urban tree managers that are not captured here (Martin et al., 2024). There is also information in textbooks and other materials that can help capture data summarized in myriad papers; however, without the original data sources we cannot assess how often the research described the socio-ecological conditions in which the work was conducted. Thus, it is believed the results in this paper provide a suitable reflection of the research intent and questions.

Furthermore, given the wealth of expertise in the urban tree care professional and practitioner communities, we recommend further research that explores expert opinion on the impacts of stewardship and maintenance practices, and the role that different contextual factors play in mediating those outcomes.

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#### CRediT authorship contribution statement

**Renata Poulton Kamakura:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Funding acquisition. **Chloe Schueller:** Writing – review & editing, Methodology, Investigation. **Brittany Wofford:** Writing – review & editing, Methodology, Investigation. **Clare E. Kazanski:** Writing – review & editing, Methodology, Conceptualization. **Richard Hallett:** Writing – review & editing, Methodology. **Richard Hauer:** Writing – review & editing, Methodology. **Rachel V. Holmes:** Writing – review & editing, Methodology, Conceptualization.



## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.ufug.2025.129121](https://doi.org/10.1016/j.ufug.2025.129121).

## References

- Allen, K.S., Harper, R.W., Bayer, A., Brazee, N.J., 2017. A review of nursery production systems and their influence on urban tree survival. *Urban For. Urban Green.* 21, 183–191. <https://doi.org/10.1016/j.ufug.2016.12.002>.
- Almas, A., Conway, T., 2018. Resident attitudes and actions toward native tree species: a case study of residents in four Southern Ontario municipalities. *Arboric. Urban For.* 44 (2), 101–115. <https://doi.org/10.48044/JAUF.2018.009>.
- Arron, G.P., Becker, S. de, Stubbs, H.A., Szeto, E.W., 1997. An evaluation of the efficacy of tree growth regulators paclobutrazol, flurprimidol, dikegulac, and uniconazole for utility line clearance. *J. Arboric.* 23, 8–16.
- Ball, John, 1992. Response of the bronze birch borer to pruning wounds on paper birch. *Arboric. Urban For. (AUF)* 18 (6), 294–297. <https://doi.org/10.48044/JAUF.1992.056>.
- Barger, J.H., Cannon Jr., W.N., 1987. Response of smaller European elm bark beetles to pruning wounds on American elm. *J. Arboric.* 13 (4), 102–104, 13: 102–4. <https://research.fs.usda.gov/treesearch/53867>.
- Blair, S.A., Koeser, A.K., Knox, G.W., Roman, L.A., Thetford, M., Hilbert, D.R., 2019. Health and establishment of highway plantings in Florida (United States). *Urban For. Urban Green.* 43, 126384–126384.
- Boyce, S., 2010. It takes a stewardship village: effect of volunteer tree stewardship on urban street tree mortality rates. *Cities Environ.* 3, 1–8.
- Braga, R.M., Souza, M.M. de, Braga, F. de A., Venturin, N., 2020. Post-pruning growth of some afforestation species in the metropolitan region of Belo Horizonte, Minas Gerais state, Brazil. *Rev. Árvore* 44. <https://doi.org/10.1590/1806-908820200000011>.
- Breger, B.S., Eisenman, T.S., Kremer, M.E., Roman, L.A., Martin, D.G., Rogan, J., 2019. Urban tree survival and stewardship in a state-managed planting initiative: a case study in Holyoke, Massachusetts. *Urban For. Urban Green.* 43, 126382. <https://doi.org/10.1016/j.ufug.2019.126382>.
- Breyer, B., Mohr, H., 2023. Right tree, right place for whom? Environmental justice and practices of urban forest assessment. *Local Environ.* 0, 1–15. <https://doi.org/10.1080/13549839.2023.2184784>.
- Brienen, R.J.W., Caldwell, L., Duchesne, L., Voelker, S., Barichivich, J., Baliva, M., Ceccantini, G., et al., 2020. Forest carbon sink neutralized by pervasive Growth-Lifespan Trade-Offs. *Nat. Commun.* 11 (1), 4241. <https://doi.org/10.1038/s41467-020-17966-z>.
- Bühler, O., Nielsen, C.N., Kristoffersen, P., 2006. Growth and phenology of established *Tilia cordata* street trees in response to different irrigation regimes. *Arboric. Urban For.* 32, 3–9.
- Cannon Jr., W.N., Worley, D.P., 1976. Dutch elm disease control: performance and costs. *USDA Serv. Res. Pap. NE345*.
- Chase, Kevin D., LeBrun, Elden, Rigsby, Chad M., 2021. Efficacy of flupyradifurone, pyriproxyfen and horticultural oil, and dinotefuran against gloomy scale (*Melanaspis tenebricosa* Comstock). *Arboric. Urban For. (AUF)* 47 (2), 64–71. <https://doi.org/10.48044/JAUF.2021.006>.
- Clark, Jeffrey J., Knudsen, Andrew C., 2013. Extent, characterization, and sources of soil lead contamination in Small-Urban residential neighborhoods. *J. Environ. Qual.* 42 (5), 1498–1506. <https://doi.org/10.2134/jeq2013.03.0100>.
- Clark, J.R., Matheny, N., 2010. The research foundation to tree pruning: a review of the literature. *Arboric. Urban For. (AUF)* 36 (3), 110–120. <https://doi.org/10.48044/jauf.2010.015>.
- Coombes, A., Martin, J., Slater, D., 2019. Defining the allometry of stem and crown diameter of urban trees. *Urban For. Urban Green.* 44 (N.PAG-N.PAG).
- Covidence systematic review software, 2023.
- Clegg, B., Ellison-Smith, D., 2020. Application of paclobutrazol to mitigate environmental stress of urban street trees. *Forests* 11. <https://doi.org/10.3390/f11030355>.
- Czaja, M., Kolton, A., Muras, P., 2020. The complex issue of urban Trees—Stress factor accumulation and ecological service possibilities. *Forests* 11, 932. <https://doi.org/10.3390/f11090932>.
- Doccola, Joseph J., Smitley, David R., Davis, Terrance W., Aiken, John J., Wild, Peter M., 2011. Tree wound responses following systemic insecticide trunk injection treatments in Green ash (*Fraxinus pennsylvanica* Marsh.) as determined by destructive autopsy. *Arboric. Urban For. (AUF)* 37 (1), 6–12. <https://doi.org/10.48044/JAUF.2011.002>.
- Faccoli, M., Gatto, P., 2016. Analysis of costs and benefits of Asian longhorned beetle eradication in Italy. *Forestry* 89, 301–309. <https://doi.org/10.1093/forestry/cpv041>.
- Fahrner, S.J., Abrahamson, M., Venette, R.C., Aukema, B.H., 2017. Strategic removal of host trees in isolated, satellite infestations of emerald ash borer can reduce population growth. *Urban For. Urban Green.* 24, 184–194. <https://doi.org/10.1016/j.ufug.2017.03.017>.
- Ferrini, F., Baietto, M., 2006. Response to fertilization of different tree species in the urban environment. *Arboric. Urban For.* 32. <https://doi.org/10.48044/jauf.2006.012>.
- Ferrini, F., & Fini, A., 2013. Trees and Management: Advances in Research. International Society of Arboriculture Annual Convention, Toronto, Canada.
- Fini, A., Degl'Innocenti, C., Ferrini, F., 2016. Effect of mulching with compost on growth and physiology of *Ulmus* "FL634" planted in an urban park. *Arboric. Urban For.* 42, 192–200.
- Follett, M., Nock, C.A., Buteau, C., Messier, C., 2016. Testing a new approach to quantify growth responses to pruning among three temperate tree species. *Arboric. Urban For.* 42, 133–145.
- Foo, Y.Z., O'Dea, R.E., Koricheva, J., Nakagawa, S., Lagisz, M., 2021. A practical guide to question formation, systematic searching and study screening for literature reviews in ecology and evolution Boolean, meta-analysis, narrative review, screening, systematic review, systematic search. *Methods Ecol. Evol.* 12, 1705. <https://doi.org/10.1111/2041-210X.13654>.
- Fraedrich, S.W., Ham, D.L., 1982. Wood chip mulching around maples: effect on tree growth and soil characteristics. *J. Arboric.* 8, 85–89.
- Funk, R., 1988. Davey's Plant health care. *Arboric. Urban For. (AUF)* 14, 285–287. <https://doi.org/10.48044/jauf.1988.069>.
- Funk, R., Peterson, W., 1980. Nutrient treatments for sugar maple decline. *J. arb* 6, 124–129.
- Ghani, M.Abd, Stokes, A., Fourcaud, T., 2009. The effect of root architecture and root loss through trenching on the anchorage of tropical urban trees (*Eugenia grandis* Wight). *Trees* 23, 197–209. <https://doi.org/10.1007/s00468-008-0269-9>.
- Gill, S., Davidson, J.A., Raupp, M.J., 1992. Control of peachtree borer using entomopathogenic nematodes. *J. Arboric.* 18, 184–187.
- Gilman, E.F., 2006. Deflecting roots near sidewalks. *Arboric. Urban For.* 32, 18–23.
- Goodfellow, John W., Blumreich, Bing, Nowacki, Greg, 1987. Tree growth response to line clearance pruning. *Arboric. Urban For. (AUF)* 13 (8), 196–200. <https://doi.org/10.48044/JAUF.1987.042>.
- Green, T.L., Watson, G.W., 1989. Effects of turfgrass and mulch on the establishment and growth of bare-root sugar maples. *J. Arboric.* 15, 268–272.
- Gregory, G.F., Allison, J.R., 1979. The comparative effectiveness of pruning versus pruning plus injection of trunk and/or limb for therapy of Dutch elm disease in American elms. *J. Arboric.* 5, 1–4.
- Großmann, J., Pyttel, P., Bauhus, J., Lecigne, B., Messier, C., 2020. The benefits of tree wounds: microhabitat development in urban trees as affected by intensive tree maintenance. *Urban For. Urban Green.* 55. <https://doi.org/10.1016/j.ufug.2020.126817>.
- Hargrave, J., Rebecca, Harper, Richard W., Butler, Brett J., Mullins, Jamie T., 2023. "Municipal forest program management in the United States of America: a systematic review". *Forests* 14 (1), 35. <https://doi.org/10.3390/f14010035>.
- Harrell, Mark O., Pierce, Philip A., Mooter, David P., 1988. Pin OAK and silver maple chlorosis treatment with ferric ammonium citrate solution. *Arboric. Urban For. (AUF)* 14 (6), 156–158. <https://doi.org/10.48044/JAUF.1988.039>.
- Harrell, Mark O., Pierce, Philip A., Mooter, David P., Webster, Bruce L., 1984. A comparison of treatments for chlorosis of pin OAK and silver maple. *Arboric. Urban For. (AUF)* 10 (9), 246–249. <https://doi.org/10.48044/JAUF.1984.053>.
- Harris, J.R., Day, S.D., Kane, B., 2008. Nitrogen fertilization during planting and establishment of the urban forest: a collection of five studies. *Urban For. Urban Green.* 7, 195–206. <https://doi.org/10.1016/J.UFUG.2008.03.001>.
- Hartman, John, Dixon, Ed, Bernick, Shawn, 2010. Evaluation of therapeutic treatments to manage oak bacterial leaf scorch. *Arboric. Urban For. (AUF)* 36 (3), 140–146. <https://doi.org/10.48044/JAUF.2010.019>.
- Hauer, R.J., Koeser, A.K., 2018. A 2018 assessment of TREE fund research grant program: research outputs, outcomes, & impacts. *TREE Research and Education Endowment Fund, Naperville, Illinois*, p. 64.
- Hauer, R.J., Miller, R.W., Werner, L.P., van den Bosch, C.C.K., 2017. The history of trees in the city. In *Routledge Handbook of Urban Forestry*. Routledge, pp. 17–32.
- Haugen, Chris, Tucker, Kevin, Smalling, Alex, Bick, Emily, Hoover, Steve, Ehlen, Grant, Watson, Todd, Bernick, Shawn, 2016. The efficacy of paclobutrazol soil application as it relates to the timing of utility Right-of-Way pruning. *Arboric. Urban For. (AUF)* 42 (2), 95–101. <https://doi.org/10.48044/JAUF.2016.008>.
- Heimpel, George E., Cock, Matthew J.W., 2018. Shifting paradigms in the history of classical biological control. *BioControl* 63 (1), 27–37. <https://doi.org/10.1007/s10526-017-9841-9>.
- Heynen, N., Perkins, H.A., Roy, P., 2006. The political ecology of uneven urban Green space: the impact of political economy on race and ethnicity in producing environmental inequality in Milwaukee. *Urban Aff. Rev.* 42, 3–25. <https://doi.org/10.1177/1078087406290729>.



- Hilbert, D., Roman, L., Koeser, A., Vogt, J., Van Doorn, N., 2019. Urban tree mortality: a literature review. *Arboric. Urban For.* 45. <https://doi.org/10.48044/jauf.2019.015>.
- Himelick, E.B., Himelick, K.J., 1980. Systemic treatment for chlorotic trees. *J. Arboric.* 6, 192–196.
- Himelick, E.B., Watson, G.W., 1990. Reduction of oak chlorosis with wood chip mulch treatments. *J. Arboric.* 16, 275–278.
- Holmes, John J., Davidson, John A., 1984. Integrated pest management for arborists: implementation of a pilot program. *Arboric. Urban For. (AUF)* 10 (3), 65–70. <https://doi.org/10.48044/JAUF.1984.011>.
- Iles, J.K., Dosmann, M.S., 1999. Effect of organic and mineral mulches on soil properties and growth of fairview flame red maple trees. *J. Arboric.* 25 (3), 163–167.
- Jack-Scott, E., Piana, M., Troxel, B., Murphy-Dunning, C., Ashton, M.S., 2013. Stewardship Success: How Community Group Dynamics Affect Urban Street Tree Survival and Growth. *Arboriculture & Urban Forestry (AUF)* 39 (4), 189–196. <https://doi.org/10.48044/JAUF.2013.025>.
- Jorgensen, E., 1986. Urban Forestry in the Rearview Mirror. *Arboricultural Journal*.
- Karlovich, D.A., Groninger, J.W., Close, D.D., 2000. Tree condition associated with topping in Southern Illinois communities. *J. Arboric.* 26, 87–91.
- Kjølgren, R.K., Clark, J.R., 1993. Growth and water relations of liquidambar styraciflua L. in an urban park and plaza. *Trees Struct. Funct.* 7, 195–201. <https://doi.org/10.1007/BF00202073>.
- Lawson, Andrew B., Dahlsten, Donald L., 2003. Implementation of a citywide monitoring program to base treatment decisions on elm leaf beetle abundance. *Arboric. Urban For. (AUF)* 29 (1), 34–41. <https://doi.org/10.48044/JAUF.2003.005>.
- Legigne, B., Delagrèze, S., Messier, C., 2018. Crown reaction and acclimation to cyclical V-trimming of city trees: an analysis using terrestrial laser scanning. *Urban For. Urban Green.* 29, 183–191.
- Legigne, B., Delagrèze, S., Messier, C., 2020. Determinants of delayed traumatic tree reiteration growth: levels of branch growth control and insights for urban tree management, modeling and future research. *Urban For. Urban Green.* 47. <https://doi.org/10.1016/j.ufug.2019.126541>.
- Lin, Jian, Wang, Qiang, Huang, Bo, 2021. Street trees and crime: what characteristics of trees and streetscapes matter. *Urban For. Urban Green.* 65. <https://doi.org/10.1016/j.ufug.2021.127366>.
- Lindén, L., Riikonen, A., Setälä, H., Yli-Pelkonen, V., 2020. Quantifying carbon stocks in urban parks under cold climate conditions. *URBAN For. URBAN Green.* 49. <https://doi.org/10.1016/j.ufug.2020.126633>.
- Livesley, S.J., McPherson, E.G., Calafapietra, C., 2016. The urban forest and ecosystem services: impacts on urban water, heat, and pollution cycles at the tree, street, and city scale. *J. Environ. Qual.* 45, 119–124. <https://doi.org/10.2134/jeq.2015.11.0567>.
- Luley, C.J., Sisinni, S., Pleninger, A., 2002. The effect of pruning on service requests, branch failures and priority maintenance in the city of Rochester, NY. *J. arboric.* 28, 137–143.
- Manfra, Rodrigo, dos Santos Massoca, Miriam, Martins Cerqueira Uras, Priscilla, Cavallari, Aline Andreia, Maselli Locosselli, Giuliano, 2022. Average height of surrounding buildings and district age are the main predictors of tree failure on the streets of São Paulo/Brazil. *Urban For. Urban Green.* 74 (August), 127665. <https://doi.org/10.1016/j.ufug.2022.127665>.
- Martin, Alexander J.F., Olson, Lukas G., Almas, Andrew D., 2024. Knowledge translation for the advancement of practice: a survey of facilitators and barriers in arboriculture and urban forestry. *Urban For. Urban Green.* 94 (April), 128242. <https://doi.org/10.1016/J.UFUG.2024.128242>.
- Massicotte P., South A., 2023. *rnaturalearth: World Map Data from Natural Earth*. R package version 1.0.1 <<https://CRAN.R-project.org/package=rnaturalearth>>.
- May, P.B., Livesley, S.J., Shears, I., 2013. Managing and monitoring tree health and soil water status during extreme drought in Melbourne, Victoria. Presented at the *Arboriculture & Urban Forestry*. International Society of Arboriculture, Champaign, USA, pp. 136–145.
- Md Meftaul, Islam, Venkateswarlu, Kadiyala, Dharmarajan, Rajarathnam, Annamalai, Prasath, Megharaj, Mallavarapu, 2020. Pesticides in the urban environment: a potential threat that knocks at the door. *Sci. Total Environ.* 711 (April), 134612. <https://doi.org/10.1016/j.scitotenv.2019.134612>.
- Morakinyo, T.E., Ouyang, W., Lau, K.K.-L., Ren, C., Ng, E., 2020. Right tree, right place (urban canyon): tree species selection approach for optimum urban heat mitigation - development and evaluation. *Sci. Total Environ.* 719, 137461. <https://doi.org/10.1016/j.scitotenv.2020.137461>.
- Morgenroth, J., 2011. Root growth response of platanus orientalis to porous pavements. *Arboric. urban for.* 37, 45–50.
- Nielsen, D.G., 1989. Integrated pest management in arboriculture: from theory to practice. *Arboric. Urban For. (AUF)* 15, 25–30. <https://doi.org/10.48044/jauf.1989.007>.
- Noack, Ann E., Kaapro, Jyri, Bartimote-Aufflick, Kathryn, Mansfield, Sarah, Rose, Harley A., 2009. Efficacy of imidacloprid in the control of thaumastocoris peregrinus on eucalyptus scoparia in Sydney, Australia. *Arboric. Urban For. (AUF)* 35 (4), 192–196. <https://doi.org/10.48044/JAUF.2009.032>.
- Noukoun, C., Bryant, G., Frank, S.D., 2014. The effect of sticky bands on cankerworm abundance and defoliation in urban trees. *Arboric. Urban For.* 40, 135–142.
- O'Herrin, K., Bassett, C.G., Day, S.D., Ries, P.D., Wiseman, P.E., 2023. Borrowed credentials and surrogate professional societies: a critical analysis of the urban forestry profession. *Arboric. Urban For. (AUF)* 49, 107–136. <https://doi.org/10.48044/jauf.2023.009>.
- Oldfield, E.E., Felson, A.J., Auyeung, D.S.N., Crowther, T.W., Sonti, N.F., Harada, Y., Maynard, D.S., Sokol, N.W., Ashton, M.S., Warren II, R.J., Hallett, R.A., Bradford, M. A., 2015. Growing the urban forest: tree performance in response to biotic and abiotic land management. *Restor. Ecol.* 23, 707–718. <https://doi.org/10.1111/rec.12230>.
- Oleksyn, J., Kloppel, B.D., Łukasiewicz, S., Karolewski, P., Reich, P.B., 2007. Ecophysiology of horse-chestnut (*Aesculus hippocastanum* L.) in degraded and restored urban sites. *Pol. J. Ecol.* 55, 245–260.
- Omar, M., Al Sayed, N., Barré, K., Halwani, J., Machon, N., 2018. Drivers of the distribution of spontaneous plant communities and species within urban tree bases. *Urban For. Urban Green.* 35, 174–191.
- Park, Sun-Jeong, Cheng, Zhiqiang, Yang, Hanbae, Morris, Elizabeth E., Sutherland, Meaghan, McSpadden Gardener, Brian B., Grewal, Parwinder S., 2010. Differences in soil chemical properties with distance to roads and age of development in urban areas. *Urban Ecosyst.* 13 (4), 483–497. <https://doi.org/10.1007/s11252-010-0130-y>.
- Pebesma, E., Bivand, R., 2023. Spatial data science: with applications in r. Chapman and Hall/CRC. <https://doi.org/10.1201/9780429459016>.
- Percival, Glynn C., 2023. Heat tolerance of urban trees – A review. *Urban For. Urban Green.* 86. <https://doi.org/10.1016/j.ufug.2023.128021>.
- Piana, M.R., Pregitzer, C.C., Hallett, R.A., 2021. Advancing management of urban forested natural areas: toward an urban silviculture? *Front. Ecol. Environ.* 19, 526–535. <https://doi.org/10.1002/fee.2389>.
- Qu, B., Liu, Y., Sun, X., Li, S., Wang, X., Xiong, K., Yun, B., Zhang, H., 2019. Effect of various mulches on soil physico-chemical properties and tree growth (*Sophora japonica*) in urban tree pits. *PLoS ONE* 14. <https://doi.org/10.1371/JOURNAL.PONE.0210777>.
- Rathjens, Richard G., Sydnor, T.Davis, Gardner, David S., 2009. Evaluating root crown excavation as a treatment for Deeply-Planted landscape trees. *Arboric. Urban For. (AUF)* 35 (6), 287–293. <https://doi.org/10.48044/JAUF.2009.043>.
- Raup, Michael J., Holmes, John J., Sadof, Clifford, Shrewsbury, Paula, Davidson, John A., 2001. Effects of cover sprays and residual pesticides on scale insects and natural enemies in urban forests. *J. Arboric.* 27 (4), 203–214. <https://doi.org/10.48044/JAUF.2001.022>.
- Raup, M.J., Shrewsbury, P.M., Gill, S., 1999. Advances in managing insect pests of urban trees. *Acta Hort.* (496), 147–154. <https://doi.org/10.17660/ActaHortic.1999.496.18>.
- Raup, Michael J., Webb, Ralph E., Szczepaniec, Adrianna, Booth, Donald, Ahern, Robert, 2004. Incidence, abundance, and severity of mites on hemlocks following applications of imidacloprid. *Arboric. Urban For. (AUF)* 30 (2), 108–113. <https://doi.org/10.48044/JAUF.2004.013>.
- Rich, S., Walton, G.S., 1979. Decline of curbside sugar maples in connecticut. *J. Arboric.* 5, 265–268.
- Richardson, W.S., Wilson, M.C., Nishikawa, J., Hayward, R.S.A. (Eds.), 1995. The well-built clinical question: a key to evidence-based decisions. *ACP J. Club* 123, A12. <https://doi.org/10.7326/ACPJC-1995-123-3-A12>.
- Riley, C.B., Gardiner, M.M., 2020. Examining the distributional equity of urban tree canopy cover and ecosystem services across United States cities. *PLOS ONE* 15, e0228499. <https://doi.org/10.1371/journal.pone.0228499>.
- Roman, L.A., Conway, T.M., Eisenman, T.S., Koeser, A.K., Ordóñez Barona, C., Locke, D. H., Jenerette, G.D., Östberg, J., Vogt, J., 2020. Beyond 'trees are good': disservices, management costs, and tradeoffs in urban forestry. *Ambio*. <https://doi.org/10.1007/s13280-020-01396-8>.
- Roman, L.A., Walker, L.A., Martineau, C.M., Muffy, D.J., MacQueen, S.A., Harris, W., 2015. Stewardship matters: case studies in establishment success of urban trees. *Urban For. Urban Green.* 14, 1174–1182. <https://doi.org/10.1016/j.ufug.2015.11.001>.
- Roseli, Ahmad Nazarudin, Mohd, Tsan, Fui Ying, Osman, Normaniza, 2021. Changes in leaf thickness, chlorophyll content, and gas exchange of a landscape tree, xanthostemon chrysanthus, treated with paclobutrazol and potassium nitrate. *Arboric. Urban For. (AUF)* 47 (2), 53–63. <https://doi.org/10.48044/JAUF.2021.005>.
- Roy, S., Byrne, J., Pickering, C., 2012. A systematic quantitative review of urban tree benefits, costs, and assessment methods across cities in different climatic zones. *Urban For. Urban Green.* 11 (4), 351–363. <https://doi.org/10.1016/j.ufug.2012.06.006>.
- Ryder, C.M., Moore, G.M., 2013. The arboricultural and economic benefits of formative pruning street trees. *Arboric. Urban For.* 39, 17–24.
- Sadof, C.S., Hughes, G.P., Witte, A.R., Peterson, D.J., Ginzel, M.D., 2017. Tools for staging and managing emerald ash borer in the urban forest. *Arboric. Urban For.* 43, 15–26.
- Sæbø, A., Ferrini, F., 2006. The use of compost in urban Green areas – a review for practical application. *Urban For. Urban Green.* 4 (3), 159–169. <https://doi.org/10.1016/j.ufug.2006.01.003>.
- Salahi, B., Hadavi, E., Samar, S.M., 2017. Foliar iron sulphate-organic acids sprays improve the performance of oriental plane tree in calcareous soil better than soil treatments. *Urban For. Urban Green.* 21, 175–182. <https://doi.org/10.1016/j.ufug.2016.12.001>.
- Sand, E., Konarska, J., Howe, A.W., Andersson-Sköld, Y., Moldan, F., Pleijel, H., Uddling, J., 2018. Effects of ground surface permeability on the growth of urban linden trees. *Urban Ecosyst.* 21, 691–696. <https://doi.org/10.1007/S11252-018-0750-1>.
- Scharenbroch, Bryant C., Fite, Kelby, Catania, Michelle, 2022. An arboriculture treatment of biochar, fertilization, and tillage improves soil organic matter and tree growth in a suburban street tree landscape in bolingbrook, Illinois, USA. *Arboric. Urban For. (AUF)* 48 (3), 203–214. <https://doi.org/10.48044/JAUF.2022.015>.
- Schultz, P.B., 1989. Forest tent caterpillar, its management as an urban pest in Virginia. *Journal of Arboriculture* 15, 92–93.
- Sherald, J.L., Gregory, G.F., 1980. Dutch elm disease therapy. *J. Arboric.* 6, 287–290.

- Shigo, A.L., Marx, H.G., 1977. Compartmentalization of Decay in Trees. Department of Agriculture, Forest Service.
- Shogren, C., Paine, T.D., 2016. Economic benefit for Cuban laurel thrips biological control. *J. Econ. Entomol.* 109, 93–99. <https://doi.org/10.1093/jee/tov283>.
- Smiley, E.T., Fraedrich, B.R., Fengler, P.H., 2000. Hazard tree inspection, evaluation, and management, in: Kuser, J.E. (Ed.), *Handbook of Urban and Community Forestry in the Northeast*. Springer US, Boston, MA, pp. 243–260. [https://doi.org/10.1007/978-1-4615-4191-2\\_17](https://doi.org/10.1007/978-1-4615-4191-2_17).
- Smiley, E. Thomas, King, Larry G., 1998. Longevity of ferric ammonium citrate treatments in oak. In: *Arboriculture & Urban Forestry (AUF)*, 24, p. 294. <https://doi.org/10.48044/JAUF.1998.036>.
- Smith, I.A., Dearborn, V.K., Hutyra, L.R., 2019. Live fast, die young: accelerated growth, mortality, and turnover in street trees. *PLoS ONE* 14, 1–17. <https://doi.org/10.1371/journal.pone.0215846>.
- Struve, D.K., 2002. A review of shade tree nitrogen fertilization research in the United States. *J. Arboric.* 28, 252–263.
- Švihra, Pavel, Burger, David, Ellis, Deborah, 1999. Effects of 3 trunk support systems on growth of young pyrus calleryana trees. *Arboric. Urban For.* 25 (6), 319–324. <https://doi.org/10.48044/JAUF.1999.042>.
- Szota, C., Coutts, A.M., Thom, J.K., Virahsawmy, H.K., Fletcher, T.D., Livesley, S.J., 2019. Street tree stormwater control measures can reduce runoff but May not benefit established trees. *Landsc. Urban Plan* 182, 144–155. <https://doi.org/10.1016/j.landurbplan.2018.10.021>.
- Takamatsu, M., Yarita, R., 2020. Repurposing a traditional Japanese method of pest control for wintering pine moths, Komo-trap, for use against summer and winter populations of fall webworms. *PeerJ* 8. <https://doi.org/10.7717/peerj.9244>.
- Tan, X., Shibata, S., 2022. Factors influencing street tree health in constrained planting spaces: evidence from kyoto city, Japan. *Urban For. Urban Green.* 67. <https://doi.org/10.1016/j.ufug.2021.127416>.
- Tate, R.L., 1980. Detection, description and treatment of girdling roots on urban Norway maple trees. University of Michigan, Ann Arbor.
- Ticehurst, Mark, Finley, Stuart, 1988. An urban forest integrated pest management program for gypsy moth: an example. *Arboric. Urban For.* 14 (7), 172–175. <https://doi.org/10.48044/JAUF.1988.042>.
- Turner, J.C.L., Buss, E.A., 2005. Biology and management of allokermes kingii (Hemiptera: Kermesidae) oak trees (*Quercus* spp. ). *J. Arboric.* 31, 198–202.
- Turner-Skoff, J.B., Cavender, N., 2019. The benefits of trees for livable and sustainable communities. *PLANTS PEOPLE PLANET* 1, 323–335. <https://doi.org/10.1002/ppp3.39>.
- Vaz Monteiro, Madalena, Levanić, Tom, Doick, Kieron J., 2017. Growth rates of common urban trees in five cities in Great Britain: a dendrochronological evaluation with an emphasis on the impact of climate. *Urban For. Urban Green.* 22 (March), 11–23. <https://doi.org/10.1016/j.ufug.2017.01.003>.
- Vogt, J., 2024. A comprehensive framework for understanding urban forests as Social-Ecological systems. *Arboric. Urban For. (AUF)* 50 (6), 427–469. <https://doi.org/10.48044/jauf.2024.021>.
- Vogt, J., Hauer, R., Fischer, B., 2015a. The costs of maintaining and not maintaining the urban forest: a review of the urban forestry and arboriculture literature. *Arboric. Urban For.* 41. <https://doi.org/10.48044/jauf.2015.027>.
- Vogt, J., Watkins, S.L., Mincey, S.K., Patterson, M.S., Fischer, B.C., 2015b. Explaining planted-tree survival and growth in urban neighborhoods: a social-ecological approach to studying recently-planted trees in Indianapolis. *Landsc. Urban Plan* 136, 130–143. <https://doi.org/10.1016/j.landurbplan.2014.11.021>.
- Watson, W.T., 2005. Influence of tree size on transplant establishment and growth. *HortTechnology* 15, 118–122. <https://doi.org/10.21273/HORTTECH.15.1.0118>.
- Watson, Gary W., Himelick, E.B., 2004. Effects of soil PH, root density, and tree growth regulator treatments on pin oak chlorosis. *Arboric. Urban For. (AUF)* 30 (3), 172–178. <https://doi.org/10.48044/JAUF.2004.021>.
- Watson, G.W., Kupkowski, G., 1991. Effects of a deep layer of mulch on the soil environment and tree root growth. *J. Arboric.* 17, 242–245.
- Werner, L.P., Jull, L.G., 2013. Fertilizer nitrogen uptake and partitioning in young and mature common hackberry (*Celtis occidentalis*) trees. *Arboric. Urban For.* 39, 85–93.
- Whitlow, T.H., Bassuk, N.L., Reichert, D.L., 1992. A 3-Year study of water relations of urban street trees. *J. Appl. Ecol.* 29, 436–450. <https://doi.org/10.2307/2404512>.
- Wickham, H., 2016. ggplot2: elegant graphics for data analysis. Springer-Verlag New York.
- Zhu, J., Cao, Y., He, W., Xu, Q., Zhang, X., 2021. Leaf functional traits differentiation in relation to covering materials of urban tree pits. *BMC Plant Biol.* 21. <https://doi.org/10.1186/s12870-021-03316-8>.